RAJALAKSHMI ENGINEERING COLLEGE

LATHA A 241701024 <latha.a.2024.csd@rajalakshmi.edu.in>

6th OPJU International Technology Conference (OTCON 6.0) on SMARTICON (Smart & Intelligent Computing for Next-Generation Technologies) : Submission (330) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org> Sun, Nov 1, 2026 at 8:23 PM
Reply-To: Microsoft CMT - Do Not Reply <noreply@msr-cmt.org>
To: latha.a.2024.csd@rajalakshmi.edu.in

Hello,

The following submission has been created.

Track Name: Smart Computing for Data Science, Big Data & Analytics

Paper ID: 330

Paper Title: EATALY: A Smart Cafeteria Management System with Intelligent Load Balancing and Real-Time Crowd Analytics Using Flutter and Firebase

Abstract:
The increasing demand for efficient food service in educational institutions and workplaces has highlighted challenges such as long queues, overcrowding, uneven counter utilization, and inefficient order management. This paper presents EATALY, an intelligent smart cafeteria management system that integrates Flutter, Firebase, and data-driven analytics to enhance operational efficiency and user experience. The system incorporates Smart Load Balancing to dynamically distribute orders based on queue length, waiting time, and counter availability, while a Real-Time Crowd Heatmap visualizes congestion levels to assist users in selecting less crowded service areas. Additional features include Digital Token Tracking, Pickup Slot Booking, Bilingual Staff Management, Secure Authentication, Real-Time Order Synchronization, Push Notifications, Order History, and an Administrative Dashboard for monitoring cafeteria operations. Experimental evaluation using simulated scenarios demonstrated reliable cloud synchronization, effective queue distribution, and improved crowd management. The proposed architecture is scalable, cost-effective, and suitable for deployment in educational institutions, corporate campuses, hospitals, and other high-demand food service environments. Future enhancements include AI-based demand forecasting, personalized recommendations, IoT-enabled occupancy sensing, and predictive analytics to further optimize cafeteria operations.

Created on: Sun, 01 Nov 2026 14:53:22 GMT

Last Modified: Sun, 01 Nov 2026 14:53:22 GMT

Authors:
- latha.a.2024.csd@rajalakshmi.edu.in (Primary)
- aishwarya.b.2024.csd@rajalakshmi.edu.in

Secondary Subject Areas: Not Entered

Submission Files:
Eataly Paper Publication.pdf (972 Kb, Sun, 01 Nov 2026 14:52:36 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our Privacy Statement.

Microsoft Corporation
One Microsoft Way
Redmond, WA 98052